

- 7 (a) *Photoelectric effect* refers to the emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency is incident on it.

Observations from photoelectric effect:

- No electrons are emitted if the frequency of the radiation is below **a minimum threshold frequency f_0** , even with very intense radiation. Above f_0 , electrons are emitted even with low-intensity radiation.
- Increasing the intensity of the radiation increases the rate at which electrons are emitted, but has no effect on the maximum energy of the electrons. The **maximum kinetic energy** of the emitted electrons **increases with the frequency** of the radiation.
- There is **no noticeable time lag** between the arrival of the radiation and the emission of the photoelectrons.

These facts imply that electromagnetic radiation arrives as discrete packets of energy known as photons, which interact with the electrons on a one-to-one basis. The photon's energy E is proportional to the radiation frequency f ($E = hf$) and not proportional to intensity. If this energy exceeds the metal's work function, it will be totally acquired by the electron instantaneously, and the photoelectron will be emitted from the metal immediately. This is the particulate nature of electromagnetic radiation.

- (b) The equation is: $hf = \phi + KE_{max}$

h is the Planck constant (6.63×10^{-34} J s),

f is the frequency of the incident radiation,

hf represents the energy of the incident photon,

ϕ is the work function of the metal, i.e. the minimum energy of the incident photon required for photoemission to take place,

KE_{max} is the kinetic energy of the most energetic photoelectron emitted from the metal surface.

- (c) Photon energy $E = hf$

$$\begin{aligned} &= \frac{hc}{\lambda} \\ &= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(3.82 \times 10^{-7})} \\ &= 5.2068 \times 10^{-19} \text{ J} \end{aligned}$$

$$\begin{aligned}
 \text{Maximum kinetic energy, } KE_{\max} &= \frac{1}{2}mv^2 \\
 &= \frac{1}{2} (9.11 \times 10^{-31})(6.87 \times 10^5)^2 \\
 &= 2.1498 \times 10^{-19} \text{ J.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Work function energy } \phi &= hf - KE_{\max} \\
 &= 5.2068 \times 10^{-19} - 2.1498 \times 10^{-19} \\
 &= 3.06 \times 10^{-19} \text{ J}
 \end{aligned}$$

(d) (i) Length = speed $\times$ time = $c t$

$$\begin{aligned}
 &= (3.00 \times 10^8)(1.0 \times 10^{-5}) \\
 &= 3.0 \times 10^3 \text{ m}
 \end{aligned}$$

(ii) Uncertainty in position $\Delta x = 3.0 \times 10^3 \text{ m}$

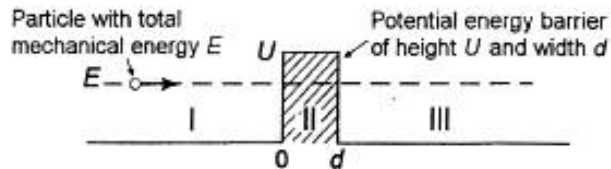
(iii) Apply Heisenberg's uncertainty principle, $\Delta x \Delta p > \frac{h}{4\pi}$

$$\Delta p > \frac{h}{4\pi(\Delta x)}$$

$$\Delta p > 1.8 \times 10^{-38} \text{ kg m s}^{-1}$$

The uncertainty in momentum is at least $1.8 \times 10^{-38} \text{ kg m s}^{-1}$

(e) A potential barrier is a region within which the potential energy U of a particle is higher than its total energy E .



Classically, a particle with total energy E lower than U will never be able to penetrate through the barrier.

However, in quantum mechanics, particles are treated as waves with a wave function ψ . The square of the wave function $|\psi|^2$, gives the probability of finding the particle at a certain position. When the wave function of the particle encounters a potential barrier, its amplitude does not become zero abruptly but decreases exponentially within the barrier, and is still non-zero beyond the barrier. Hence there is a non-zero probability that a particle can tunnel through a barrier.

